1: Linear system solver

In this part we created a function, `tridiagonal_solve()` that solves the linear system of equations by defining the problem such that the known variables are defined in a tri-diagonal matrix system, and the unknowns are defined by the vector of w_ν elements:

$$\begin{bmatrix} 1+2\lambda & -\lambda & 0 & \cdots & 0 \\ -\lambda & 1+2\lambda & -\lambda & \ddots & \vdots \\ 0 & -\lambda & 1+2\lambda & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & -\lambda \\ 0 & \cdots & 0 & -\lambda & 1+2\lambda \end{bmatrix} \begin{bmatrix} w_{1,\nu+1} \\ w_{2,\nu+1} \\ w_{3,\nu+1} \\ \vdots \\ w_{M,\nu+1} \end{bmatrix} = \begin{bmatrix} w_{1,\nu} \\ w_{2,\nu} \\ w_{3,\nu} \\ \vdots \\ w_{M,\nu} \end{bmatrix} + \begin{bmatrix} \lambda w_{0,\nu+1} \\ 0 \\ \vdots \\ 0 \\ \lambda w_{M+1,\nu+1} \end{bmatrix} \tag{1}$$

The system of equations shown in (1) are generated using the **implicit method**, and are written in matrix-vector form as $Aw_{\nu+1} = w_\nu + d_\nu$. d_ν is the vector for the boundry condition, and w_ν corresponds to the vector from the dependent variable y in the heat equation. Using the **Thomas Algorithm** the system of equations with a tri-diagonal matrix as shown in (1) can be solved.

To test this function works in python we tested a simple case for $\lambda = 1$ and $w_\nu = [4, 6, 8]$, such a system of equations can easily be solved by hand, the result was $[2.76, 4.29, 4.10]$.

2: Bounded power put option boundary conditions

In this part we will investigate the boundary conditions of a bounded power put option:

$$h(s) = \min(L, ((K - s)^+)^p) \tag{2}$$

Using the Black scholes partial differential equation (PDE): $\frac{\partial V(s,t)}{\partial t} + rs\frac{\partial V(s,t)}{\partial s} + \frac{1}{2}\sigma^2 s^2 \frac{\partial^2 V(s,t)}{\partial s^2} - rV(s,t) = 0$ and the heat equation: $\frac{\partial y}{\partial \tau} - \frac{\partial^2 y}{\partial x^2} = 0$. Because both have different dependent and independent variables the boundry conditions are also different, the boundary conditions for the bounded power option given under the Black-Scholes PDE are given below:

$$\lim_{s\to 0} V(s,t) = Le^{-r(T-t)}, \text{ for all } t \in [0,T], \text{ when } p \neq 1$$

$$\lim_{s\to \infty} V(s,t) = 0, \text{ for all } t \in [0,T]$$

$$V(s,t) = e^{-r(T-t)} h(s), \text{ for all } s > 0$$

The first boundary condition is evaluated by recognising that as $s \to 0$ the payoff is $\min(L, K^p)$ and given that $L < K^p$ when $p \neq 1$, the payoff for $s \to 0$ is L. Given that the value of the option is the **expected value** of the **discounted payoff**, $V(s,t) = e^{-r(T-t)}\mathbb{E}\left(h(S_T) \mid S_t = s\right)$ the value of the option at the first boundary condition is derived as $Le^{-r(T-t)}$, under risk-neutral valuation. The second boundary condition, for $s \to \infty$ is simplified as the payout for a regular put option as $s \to \infty$, which is 0.

The Black-Scholes PDE can be transformed into the heat equation by changing the variables (s,t) into (x,τ), using the relationship shown in (6.7) of the lecture notes. The change of variables gives the heat equation $\frac{\partial y}{\partial \tau} - \frac{\partial^2 y}{\partial x^2} = 0$. Under this new relationship we can establish new boundary conditions for the new variables, at the limits $x \to \infty$ and $x \to -\infty$, the generalised form for the boundary conditions are:

$$\lim_{x\to-\infty} y(x,\tau) = e^{-r(T-\frac{2\tau}{\sigma^2})} r_1 \left(T - \frac{2\tau}{\sigma^2}\right), \text{ for } \tau \in \left[0, \frac{1}{2}\sigma^2 T\right],$$

$$\lim_{x\to\infty} y(x,\tau) = e^{-r(T-\frac{2\tau}{\sigma^2})} r_2 \left(T - \frac{2\tau}{\sigma^2}\right), \text{ for } \tau \in \left[0, \frac{1}{2}\sigma^2 T\right]$$

r_1 and r_2 are generic terms, but for the bounded power put option we can pick specific values for them based on the boudaries used for the Black-Scholes PDE. Since $y(x,\tau)$ and $V(s,t)$ are proportional, the second limit is the same as the lower limit for the Black-Scholes PDE: 0. Similarly because y and V are proportional we can assume at the other extreme limit the two behave similarly. Recognising that $t = \frac{2\tau}{\sigma^2}$, shows the similarilty between the Black-Scholes lower limit and the heat equation lower limit, we can infer from this that $r_1\left(T - \frac{2\tau}{\sigma^2}\right)$ is equal to the payoff as $x \to -\infty$, which is $\min(L, K^p) = L$ for the same reason explained before. The final limits for the heat equation are then:

$$\lim_{x\to-\infty} y(x,\tau) = Le^{-r(T-\frac{2\tau}{\sigma^2})}, \text{ for } \tau \in \left[0, \frac{1}{2}\sigma^2 T\right], \text{ when } p \neq 1$$

$$\lim_{x\to\infty} y(x,\tau) = 0, \text{ for } \tau \in \left[0, \frac{1}{2}\sigma^2 T\right]$$

$$y(x,\tau) = e^{-r\left(T-\frac{2\tau}{\sigma^2}\right)} h(s), \text{ for all } x \in \mathbb{R}$$

3: Numerical solution of the Black-Scholes PDE Implicit method

In this part we implemented the function `bounded_power_implicit()` to calculate the option prince at the $(M+1) \times (N+1)$ grid points alongside the stock prices and time vector for a vanilla European put option. The algorithm used was based off algorithm 7.2 in the lecture notes. Firstly the values for $\partial\tau$, ∂x and λ were implemented. Then the payoffs for the i number of stock prices at $\tau = 0$ were found, $w_{i,0}$. The heat equation transformation: $S = e^{x-\left(\frac{2r}{\sigma^2}-1\right)\tau}$ was used when $\tau = 0$ which then simplifies to $S = e^x$.

The boundary conditions for the bounded power put option were then implemented but for $S \to 0$ and for $S \to \infty$, for $S \to \infty$. This sets up the grid such that across all times these values are fixed, as $S \to 0$, $V = e^{-rt}h(s)$ and as $S \to \infty$, $V = 0$. Finally, the right hand side of equation (1) was set up, by adding the boundary conditions to the $w_{i,\nu}$ array as is shown in (1) again, which gives $w_\nu + d_\nu$ in vector form. The $w_{i+1,\nu}$ vector was then returned alongside the stock price and time grid.

We then computed the Black-Scholes analytical solution, to the price of a European put $Ke^{-rT}\Phi(-d_2) - S_0\Phi(-d_1)$ and compared its value to the implicit method option price of a European put (with parameters $p = 1$ and $L = \infty$). The vanilla European put option prices for time $t = 0$ were then computed for a range of stock prices, using the lower limit `xmin = -10`. This limit was chosen for $\tau = 0$ as this corresponds to a stock price of e^{-10}, much lower than this, and we would have to increase the number of spatial stock price points on the grid, this number gave an accurate plot for the behaviour of the European put option using `M = 1000` and `N = 1000`. The plot for the analytical and numerical solution of the vanilla European put option is shown below:

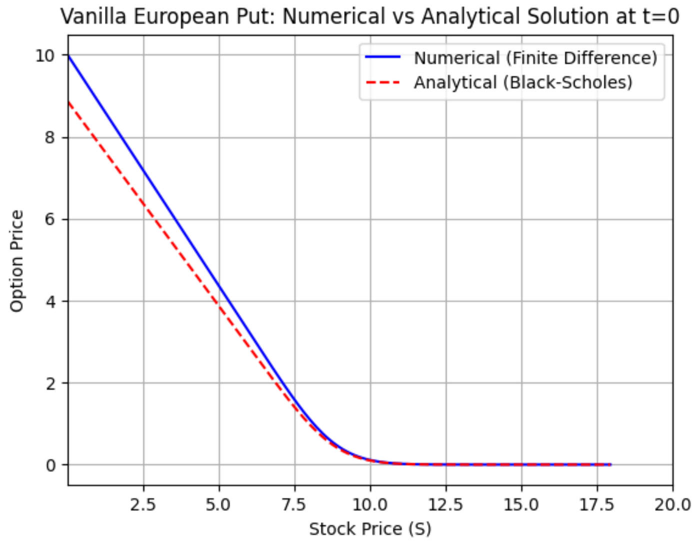

Figure 1: Plot of the numerical (implicit method) and analytical solutions for the a vanilla European put option.

Generating the plot for Figure 1 required finding a suitable value for M, to make the predictions accurate enough without drastically increasing computational time. For this part we will analyse how M affects the round off error when fixing the time-grid resolution at N = 1000. The process when solving for $w_{\nu+1}$ by hand requires simple matrix manipulation: $w_{\nu+1} = A^{-1}(w_\nu + d_\nu)$, unfortunately computers cannot calculate the solution in exactly this way, therefore an error is introduced. The cumulative error generated by the computer is $A^{-v}\mathbf{r}_0 + A^{-(v-1)}\mathbf{r}_1 + \cdots + A^{-1}\mathbf{r}_{v-1} + \mathbf{r}_n\mathbf{u}$. If the modulus of the eigenvalue of $A^{-1} < 1$ the total round-off error will be reduced upon each iteration, conversely if the modulus of the eigenvalue of $A^{-1} > 1$ the total round-off error will increase upon each iteration.

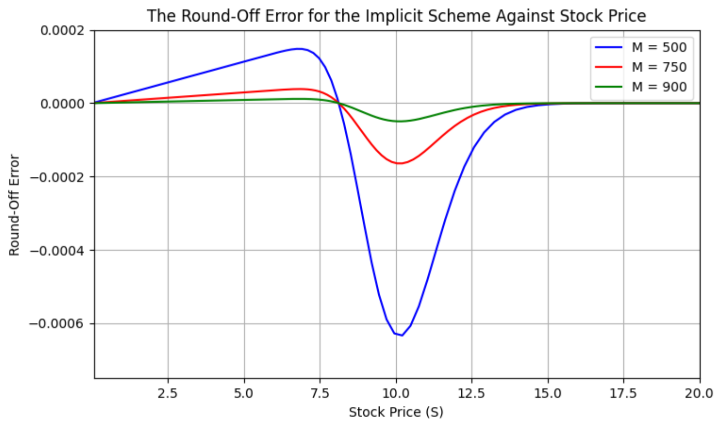

Figure 2: Plot of the round off error for the implicit scheme at M values of 500, 750 and 1000.

The plot above shows that increasing M reduces the round-off error generated by the computer which suggests that the modulus of the eigenvalue of A^{-1} is less than one so increasing the number of iterations reduces the overall round-off error. The errors also peaked around $S = 10$ which is the strike price for the put option. The errors peak around this value because the option value begins to change non-linearly and approaches zero and this breaks continuity in the payoff. The discontinuity of the change in the option price and the stock price can be visualised in Figure 1 for M = 500 there are visible 'kinks' in the plot as the stock price approaches the strike price.

4: Analysis of the case p = 2

In this part we will investivate the value of the bounded power put option, using the parameters `p = 2` and `L = 81`. from the implementation of the function: `bounded_power_implicit()`, we generated the grid values for the stock price, times and the option values, which were used to create a surface plot for the option:

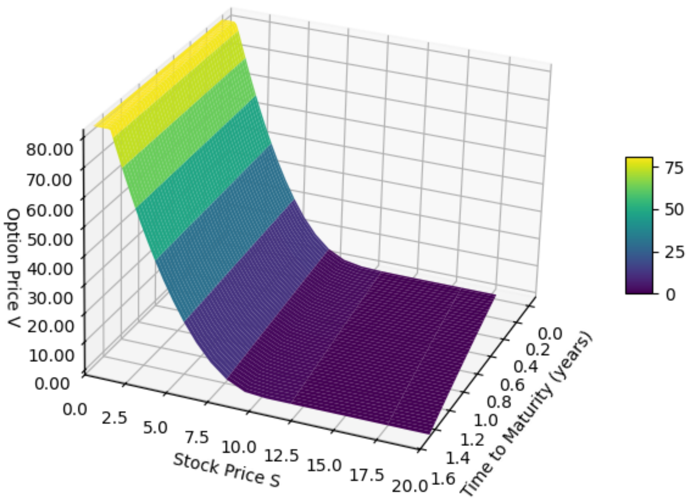

Figure 3: 3D Surface plot of the measures that influence the option value (p = 2).

The plot shows the bouded nature of the option at low stock prices, where the option is heavily in the money the payoff is capped at 81, then as the stock price increases the option price becomes more out of the money. The option value exhibits time dependence with the option value as expected, but also with the stock price. This is because of the heat equation transformation for stock price $S = e^{x-\left(\frac{2r}{\sigma^2}-1\right)\tau}$. The effect of this transformation is that trajectories that have constant stock price but decreasing time to maturity exhibit **decrease in option value with time.**

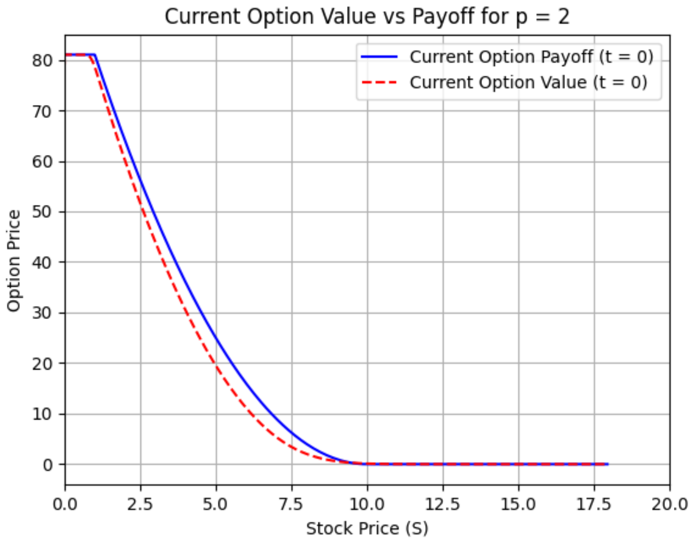

Figure 4: Current Option value vs its payoff at t = 0 for p = 2 using the implicit method.

The payoff exhibits a similar trend to the option value generated by the implicit scheme at t = 0, however the payoff within the limits of the boundary conditions is always higher than the option price which is expected. If we plotted the same for the terminal time we would see that the option value becomes the option's payoff.

In this following part we will show the use of interpolation to find the option price at the stock price €4.73 and then show how it can be used to estimate the greek, Theta. Using interpolation to find the option price at the exact stock price of €4.73 produced the value **€22.21** from the implicit numerical method. This value is in agreement with the plot shown in figure 4 for the option value against stock price at the current time.

The time dependency of an option's value can be quantified using the theta of an option: $\Theta = \frac{\partial V}{\partial t}$, using the forward differencing approximation $\frac{df}{dx} \approx \frac{f(x+\Delta x)-f(x)}{\Delta x}$ we can calculate an approximation for this. However theta **assumes that the stock prices are constant** through this difference. However the grid generated by the function clearly shows that the stock prices are not constant across the time step, this is problematic for the value of theta that will be predicted. To fix this we need to interpolate using a fixed array of stock prices for both.

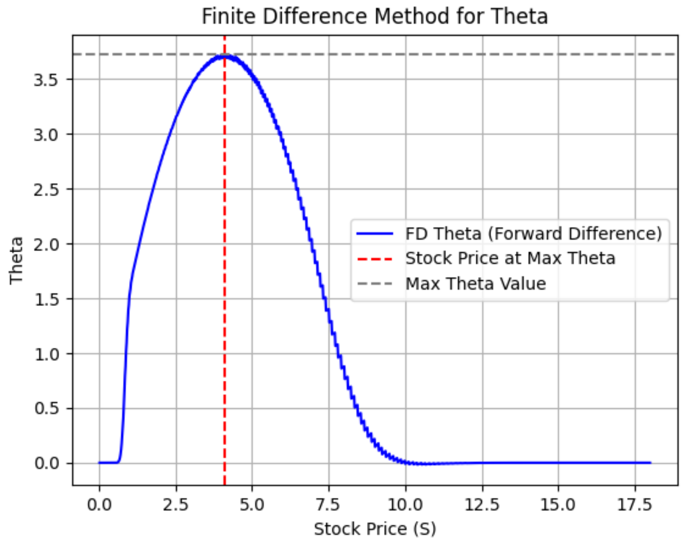

Figure 5: Range of theta values calculated by the finite difference method, with the maximum shown by the horizontal and vertical lines

Above shows the range of theta values generated from the finite difference method of the implicit scheme, the maximum stock price value for theta was **€4.12**, and the maximum theta value was **3.72.** The visible kinks shown by the plot is a result of the amount of timesteps chosen, and the use of interpolation to estimate theta for a consistent set of stock prices. The value of M and N were chosen to be 1000 which produced accurate estimations whilst not compromising the computational time.

5: Analysis of the case p = 0.5

In this part we will investivate the value of the bounded power put option, using the parameters `p = 0.5` and `L = 3`. The grid values for the stock price, times and the option values, were computed and used to create a surface plot for the option:

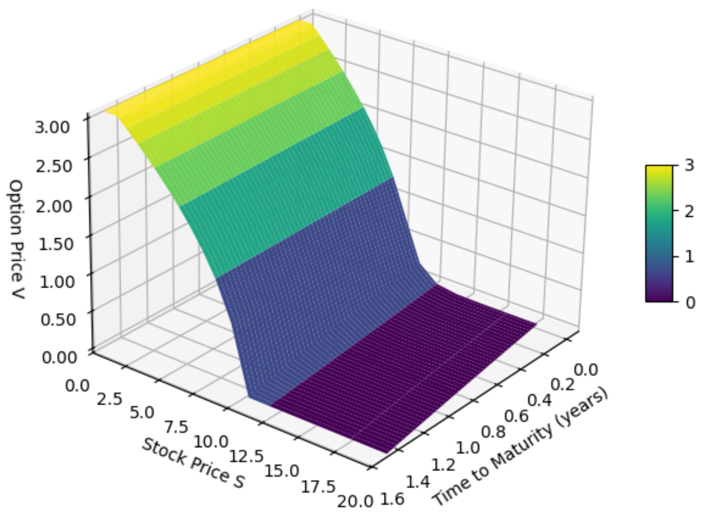

Figure 6: 3D Surface plot of the measures that influence the option value (p = 0.5).

The plot above shows the bouded nature of the option at low stock prices, where the option is heavily in the money the payoff is capped at 3. As the stock price increases the option becomes more out of the money; the option value drops off slowly further away from the strike price and accelerates out of the money much faster than the p = 2 plot when it approaches the strike price. This is the effect of the power term being less than one. We notice a similar time dependency on the option price in this plot, where at a fixed stock price, the **the option value decreases with time.**

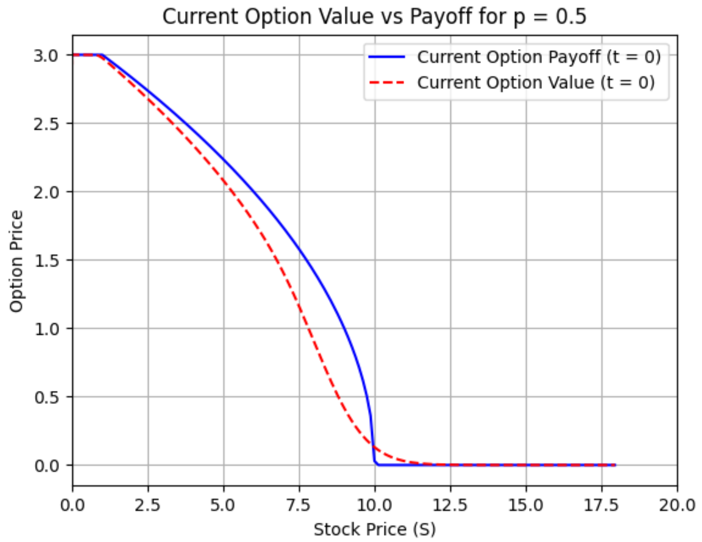

Figure 7: Current Option value vs its payoff at t = 0 for p = 0.5 using the implicit method.

The payoff exhibits a similar trend to the option value generated by the implicit scheme at t = 0, however the payoff within the limits of the boundary conditions is higher than the option price at all points except just beyond the strike price. This is a result of the payout shape generated by the power term being 0.5, which causes a sharp change in the payout about the strike price.

In this following part we will repeat the method of interpolation to find the option price at the stock price €4.73 and then show how it can be used to estimate theta. Using interpolation to find the option price at the exact stock price of €4.73 produced the value **€2.15** from the implicit numerical method. This value is in agreement with the plot shown in figure 7 for the option value against stock price at the current time.

Calculating the maximum value for theta in the same way as part 4 gave the maximum value at **€7.81**, and the maximum theta value was **0.34**. The value of M and N were again chosen to be 1000 which produced accurate estimations whilst not compromising the computational time.

6: Results and Discussion

The aim of this report was to use the implicit method to solve the price of a bounded power put option. The method adopted in this report used the Thomas Algorithm to solve a system of equations defined within known boundary conditions. Then by defining a set of grid points withing spatial and time boundaries, the unknowns in the system of equations could be found.

To implement this method the boundary conditions must first be set for the bounded power put option, such conditions were first found for the Black-Scholes PDE, by simplifying the problem one could recognise the constraint on $L < K^p$ and the through the risk neutral valuation of the discounted payoff its simplified boundary conditions were found. Applying the heat equation transformation yielded the corresponding boundary conditions for the transformed variables.

The implicit scheme for solving the price of a vanilla European put option (using parameters $p = 1$ and $L = \infty$) was compared to the Black-Scholes price of the corresponding option using the general analytical solution which suggested that the implicit schem worked well as a numerical prediction for the price. For this part the parameters chosen were: `xmin = -10`, `xmax = 3`, `N = 1000`, `M = 1000`, the lower limit for x was chosen so that the stock price was close enough to zero ($e^-10 \approx 0$) and the upper limit such that the stock price was close to 20 ($e^3 = 20.08$). The values for N and M were chosen to produce accurate plots with smooth curvature around the strike price, without comprimising the computational time.

The next part involved justifying the optimal number of spatial parameters M that were used throughout. It was found that at a fixed time grid resolution (N = 1000) there was a round off error induced by the change in the number of M. At lower values of M, visible kinks created by discontinuity in the change of the option price about the strike price were shown, and a larger round off error was induced. Increasing M to 1000 solved this issue.

For part 4 the grid values for option price stock price and time to maturity were compiled into a 3D surface plot for the parameters p = 2 and L = 81, visible time-dependent trends were shown with the option and stock price which were explained by the heat equation transformation. Then by plotting the option value against its payoff at t = 0, a visible difference was shown between the option value and its payoff at this time. The time dependence aspect can be shown if we produce the same plot at the terminal time, we would observe the option value converge to its payoff. By using interpolation, we then found a value for the option price at the current stock price, €4.73 which agreed with the plot shown in figure 4. A similar interpolation method was adopted to produce a maximum value of theta, along with a plot that showed the change in theta over a range of stock prices, the maximum change in option value was at the stock price €4.12. The same was done for the bounded power put option with parameters p = 0.5 and L = 3 in part 5, it was found that the option had a maximum theta at a stock price of €7.81.

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